

The Cohen–Macaulay Classification for Increasing-Injection Chains of Cycle Edge Ideals

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September 6, 2026

Abstract

Let C_n be the cycle on $[n]$, let $n \geq 6$ and $r \geq 0$, and let I_{n+r} be the ideal generated by all increasing-injection images in $k[x_1, \dots, x_{n+r}]$ of the edge ideal of C_n . We prove, over every field k , that

$$k[x_1, \dots, x_{n+r}]/I_{n+r} \text{ is Cohen–Macaulay} \iff r \geq \left\lfloor \frac{n-4}{2} \right\rfloor \text{ and } r \neq n-4.$$

This resolves the cycle-chain classification posed by Anwar, Ghayas, and Javed. The proof classifies all facets of the associated interval-gap independence complex. Nonpurity handles nearly all parameters below the threshold; the sole pure exceptional family is excluded by an integral one-cocycle whose pairing with an explicit one-cycle is -1 . The remaining cases reduce to connected one-dimensional complexes, a disconnected perfect matching, or nonempty zero-dimensional complexes. In particular, no characteristic-dependent branch occurs.

1 Introduction

For a graph G on $[n]$, increasing injections from $[n]$ into a larger ordered vertex set send its edge ideal to a family of squarefree monomial ideals. Anwar, Ghayas, and Javed studied Cohen–Macaulayness for several such families [AGJ26]. For cycles they proved the complete-graph range $r \geq n-3$ and formulated the full classification appearing in the abstract as their Conjecture 3.15. We prove that classification in its stated range and over an arbitrary field.

The argument is elementary after translating the ideal into an independence complex. With $L = r+2$ and $U = n-2$, its faces are precisely the subsets whose pairwise gaps lie in $[L, U]$. We first give an exact facet criterion. It detects nonpurity throughout the low range apart from $U = 3L-1$. For that exceptional pure two-dimensional family, a short integral homology certificate works after reduction over every field. In the high range, the complex has dimension at most one and can be classified directly.

Exact integer certificates also occur in computational work on monomial algebras and independence complexes. For example, Zeng [Zen26] verifies a characteristic-zero weak Lefschetz statement for whiskered graphs on eight vertices using exact finite certificates. That finite result is methodologically related but logically independent: it neither addresses nor implies the cycle-chain classification proved here.

2 The interval-gap complex

Fix a field k , integers $n \geq 6$ and $r \geq 0$, and put $m = n + r$. Write $R_m = k[x_1, \dots, x_m]$. Let C_n have edge set

$$E(C_n) = \{\{i, i+1\} : 1 \leq i < n\} \cup \{\{1, n\}\}.$$

Let $I(C_n) \subseteq k[x_1, \dots, x_n]$ be its edge ideal. If $\text{Inc}(\mathbb{N})_{n,m}$ denotes the strictly increasing injections $\pi : [n] \rightarrow [m]$, define

$$I_{n+r} = \sum_{\pi \in \text{Inc}(\mathbb{N})_{n,m}} \pi(I(C_n))R_m, \quad \pi(x_i) = x_{\pi(i)}.$$

Thus I_{n+r} is the edge ideal of a graph $G_{n,r}$ on $[m]$.

Proposition 1 (Gap description). *For $1 \leq a < b \leq m$,*

$$\{a, b\} \in E(G_{n,r}) \iff b - a \leq r + 1 \text{ or } b - a \geq n - 1.$$

Consequently, on setting

$$L = r + 2, \quad U = n - 2,$$

the independence complex $\Delta_{n,r} = \text{Ind}(G_{n,r})$ is

$$\Delta_{n,r} = \{S \subseteq [m] : L \leq b - a \leq U \text{ for every } a < b \text{ in } S\}. \quad (2.1)$$

Moreover, $R_m/I_{n+r} = k[\Delta_{n,r}]$.

Proof. The pair $\{a, b\}$ is the image of a consecutive edge of C_n exactly when the other $n - 2$ images can be placed outside the open interval (a, b) . The number of available positions is

$$(a - 1) + (m - b) = m - (b - a) - 1,$$

so this is possible exactly when $b - a \leq m - n + 1 = r + 1$. The pair is the image of the wrap edge $\{1, n\}$ exactly when the other $n - 2$ images fit strictly between a and b , which is equivalent to $b - a \geq n - 1$. This proves the edge description. Its complement is the gap interval $r + 2 \leq b - a \leq n - 2$, giving (2.1). The final identity is the usual edge-ideal/independence-complex correspondence. $\square$

We now suppress the subscripts and regard $\Delta = \Delta_{n,r}$ as the interval-gap complex in (2.1). Notice that

$$L \geq 2, \quad U \geq 4, \quad m = L + U. \quad (2.2)$$

Lemma 2 (All facets). *Let $S = \{s_0 < s_1 < \dots < s_q\}$ be a nonempty face of Δ , put $g_i = s_i - s_{i-1}$ for $1 \leq i \leq q$, and put $D = s_q - s_0$. Then S is a facet if and only if*

$$L \leq g_i \leq 2L - 1 \quad (1 \leq i \leq q), \quad U - L + 1 \leq D \leq U. \quad (2.3)$$

Consequently, a facet of dimension $q \geq 0$ exists if and only if

$$qL \leq U, \quad q(2L - 1) \geq U - L + 1. \quad (2.4)$$

In particular, the set of facet dimensions is the integer interval

$$\max\left(0, \left\lceil \frac{U - L + 1}{2L - 1} \right\rceil\right) \leq q \leq \left\lfloor \frac{U}{L} \right\rfloor. \quad (2.5)$$

Proof. The face condition is equivalent to $g_i \geq L$ for every i and $D \leq U$. If some $g_i \geq 2L$, inserting $s_{i-1} + L$ enlarges the face. Suppose instead that $D \leq U - L$. If $s_0 > L$, then $s_0 - L \in [m]$ can be inserted. If $s_0 \leq L$, then

$$s_q + L = s_0 + D + L \leq L + U = m,$$

so $s_q + L$ can be inserted. Hence every facet satisfies (2.3).

Conversely, under (2.3), a point inserted into an internal gap would have to be at distance at least L from both adjacent vertices, forcing that gap to be at least $2L$. A point inserted outside $[s_0, s_q]$ would make the new diameter at least $D + L \geq U + 1$. Neither is possible, so S is a facet.

For a fixed q , the sums of q integers in $[L, 2L - 1]$ are exactly the integers in $[qL, q(2L - 1)]$, with the empty sum interpreted as zero. This interval meets $[U - L + 1, U]$ exactly when (2.4) holds. Taking $s_0 = 1$ realizes every such choice inside $[m]$. Formula (2.5) follows immediately. $\square$

3 Below the threshold

Lemma 3. *If $2L \leq U$, then $k[\Delta]$ is not Cohen–Macaulay over any field k .*

Proof. Write $U = QL + s$, where $Q \geq 2$ and $0 \leq s < L$. Lemma 2 always supplies a facet of dimension Q . It supplies one of dimension $Q - 1$ exactly when

$$(Q - 1)(L - 1) \geq s + 1. \quad (3.1)$$

If (3.1) holds, the complex is not pure, and hence cannot be Cohen–Macaulay. If it fails, then

$$(Q - 1)(L - 1) \leq s \leq L - 1.$$

Since $L \geq 2$ and $Q \geq 2$, this forces $Q = 2$ and $s = L - 1$. Thus the only case not already excluded by nonpurity is

$$U = 3L - 1. \quad (3.2)$$

By (2.5), Δ is then pure of dimension two.

Orient every edge increasingly and define an integral one-cochain ω by

$$\omega([a, b]) = \begin{cases} 1, & L \leq b - a \leq 2L - 1, \\ 2, & 2L \leq b - a \leq 3L - 1, \end{cases} \quad (a < b), \quad (3.3)$$

extended antisymmetrically. If $a < b < c$ is a triangle, then each consecutive gap lies in $[L, 2L - 1]$, while their sum lies in $[2L, 3L - 1]$. Hence

$$(\delta\omega)([a, b, c]) = \omega([b, c]) - \omega([a, c]) + \omega([a, b]) = 1 - 2 + 1 = 0.$$

Thus ω is a cocycle over $\mathbb{Z}$, and remains one after reduction over any field.

Consider the one-chain

$$z = [1, L + 2] - [2, L + 2] + [2, 2L + 1] - [1, 2L + 1]. \quad (3.4)$$

Its four gaps are $L + 1, L, 2L - 1, 2L$, so all four terms are edges. Directly, $\partial z = 0$, whereas

$$\langle \omega, z \rangle = 1 - 1 + 1 - 2 = -1. \quad (3.5)$$

After reduction to any field, (3.5) is still nonzero, including in characteristic two. Therefore z is not a boundary and $\tilde{H}_1(\Delta; k) \neq 0$. Reisner's criterion [Rei76] applied to the empty face now excludes Cohen–Macaulayness in the remaining two-dimensional case (3.2). $\square$

Remark 4. The exceptional equation $U = 3L - 1$ is equivalent to $n = 3r + 7$. It is the unique low-range family for which a purity obstruction alone does not settle the question.

4 At and above the threshold

Lemma 5. *Suppose $2L > U$.*

1. *If $L < U$, then Δ is a pure connected one-dimensional complex, and $k[\Delta]$ is Cohen–Macaulay over every field.*
2. *If $L = U$, then Δ is a disjoint union of L edges, and $k[\Delta]$ is not Cohen–Macaulay over any field.*
3. *If $L > U$, then Δ is a nonempty zero-dimensional complex, and $k[\Delta]$ is Cohen–Macaulay over every field.*

Proof. If $L \leq U < 2L$, Lemma 2 says that every facet is an edge. Assume first that $L < U$. For every $1 \leq i < U$, the two edges

$$\{i, i + L + 1\}, \quad \{i + 1, i + L + 1\} \quad (4.1)$$

have gaps $L + 1$ and L and join i to $i + 1$. Their displayed vertices lie in $[m]$ because $i + L + 1 \leq U + L = m$. Thus $1, \dots, U$ lie in one component. Every $j > U$ is adjacent to $j - L \in [1, U]$, so the graph is connected. A pure one-dimensional simplicial complex is Cohen–Macaulay over a field exactly when it is connected, by the one-dimensional case of Reisner's criterion.

If $L = U$, then $m = 2L$ and the only edges are $\{i, i + L\}$ for $1 \leq i \leq L$. This disconnected perfect matching is not Cohen–Macaulay. Finally, if $L > U$, no two-element face exists. The complex is nonempty and zero-dimensional, so Reisner's criterion is vacuous below its dimension and its Stanley–Reisner ring is Cohen–Macaulay over every field. $\square$

5 Classification

Theorem 6. *For every field k , every integer $n \geq 6$, and every integer $r \geq 0$,*

$$R_{n+r}/I_{n+r} \text{ is Cohen–Macaulay} \iff r \geq \left\lfloor \frac{n-4}{2} \right\rfloor \quad \text{and} \quad r \neq n-4.$$

Proof. For integral n and r ,

$$2L > U \iff r \geq \left\lfloor \frac{n-4}{2} \right\rfloor. \quad (5.1)$$

Furthermore,

$$L = U \iff r = n - 4, \quad L < U \iff r \leq n - 5, \quad L > U \iff r \geq n - 3. \quad (5.2)$$

Below the threshold in (5.1), Lemma 3 gives non-Cohen–Macaulayness over every field. Above the threshold, Lemma 5 gives Cohen–Macaulayness except in the middle case $L = U$, which is exactly the excluded value in (5.2). Proposition 1 identifies the resulting Stanley–Reisner rings with the quotients in the statement. $\square$

All obstructions in the proof are either nonpurity, disconnectedness, or the unit-valued pairing (3.5), while all positive cases use connectivity or zero-dimensionality. Hence the conclusion is uniform in the characteristic of k .

Computational and AI assistance disclosure

AI assistance was used in proof development, checking elementary parameter inequalities, and manuscript preparation. Exact finite enumerations were used during checking only as corroboration; no finite computation is invoked in the proof. The mathematical argument above is self-contained, and the integral cycle–cocycle pairing is displayed explicitly so that its behavior over every field can be checked directly.

References

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